How Much Does It Cost to Top a Multi-Task Leaderboard? Minimum Targeted Investment under Different Aggregation Rules

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Abstract

Multi-task leaderboards rank models by aggregating per-task scores, so a developer can try to put a model first by raising its scores on selected sub-datasets, for instance by training on related data. We ask how much such *targeted investment* first place requires and how this depends on the aggregation rule. We formalise it as the minimal score increase of one target model, all others fixed, under a per-task headroom cap, measured by total points (L_1), tasks touched (K) and largest single-task jump (L_∞). For score-based rules (mean, geometric and harmonic mean, median, min, trimmed mean) and rank-based rules (Borda, Condorcet-winner, plurality; Copeland exactly in 34 target–cap cases) we give exact solvers, validated against brute force, and prove closed forms where they exist; for the mean the cost is exactly n times the gap to the leader. On HELM MMLU (79 models, 57 subjects) and the Open LLM Leaderboard (4 496 models, 6 tasks) the answer is *conditional*. With headroom up to 100 points every target can be made first under every rule run with that cap, but the cost differs by up to an order of magnitude across rules (HELM, ranks 11–30: median 435 points under the mean, 38 under the median, 25 under plurality); with per-task gains capped at 10 points most mid- and low-ranked targets cannot be made first under most rules at all. The relative cost of rank-based rules does not carry over between the boards; the number of tasks, not of models, drives the divergence: on the same models, 39 leaf tasks make the median and min rules 5–20 times cheaper per task (ranks 11–100) than 6 benchmark aggregates, while the mean costs the same. The cheap rules are cheap only for optimal allocation: random spending needs 3–40 times the optimal budget under rank rules, median and min. We state the modelling assumptions (scores can be raised up to a cap, other models are static) and where they fail.

1 Introduction

Multi-task benchmarks have become the standard way to compare machine-learning models: a leaderboard consists of several sub-datasets, each model receives a score on each, and an *aggregation rule* (typically the arithmetic mean, sometimes a median or a rank-based rule) turns the score vector into one number and one ranking. Ranks influence which models are adopted, which papers are published and which products are funded, so developers have an incentive to act strategically. One concrete action is *targeted investment*: adding data related to some sub-datasets to the training of the model one wants to promote, which raises its score on those sub-datasets. The leaderboard designer chooses the aggregation rule, and the rule decides how a given amount of targeted effort translates into rank.

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This paper asks: *on public multi-task leaderboards, what targeted investment does a developer need to make a chosen target model rank first, and how does this depend on the aggregation rule?* We make the question precise (Section 3): a target model may raise its score on each task t to any value between its current score and a headroom cap c_t, all other models stay fixed, and "first" means strictly ahead of every other model under the rule. Investment is measured three ways, because they answer different questions: total score points (L_1, "how much effort"), number of tasks touched (K, "how broad"), and the largest single-task jump (L_∞, equal to the uniform improvement that would be needed if effort cannot be targeted). The headroom cap is the main modelling assumption, so we vary it systematically—from the trivial cap of 100 points to caps calibrated on observed fine-tuning gains.

Contributions.

1. **A formal model and exact solvers.** We define the minimum-investment problem for score-based (mean, geometric, harmonic, median, min, trimmed mean) and rank-based (Borda, Condorcet-winner, plurality) rules, give exact solvers (closed forms, water-filling, mixed-integer programs) and test them against brute-force enumeration.
2. **Closed-form results, checked numerically.** For the mean the minimal L_1 is exactly n times the gap to the leader (plus the margin), the minimal K is a greedy headroom count, and the min rule requires lifting every below-threshold task; we verify these identities on every target of both data sets.
3. **Empirical answer on public data.** On HELM MMLU v1.13.0 and the Open LLM Leaderboard we report the minimal investment for every target, by rank group, rule and cap, with bootstrap intervals over targets, and robustness to score normalisation, leaderboard slice, evaluation noise, task resampling and task-set size.
4. **A conditional conclusion.** Whether a leaderboard can be gamed, and how cheaply, depends on the rule *and* on the headroom assumption; we state where each of our findings holds and where it fails.

2 Related work

Benchmarks and the choice of aggregation. MMLU [Hendrycks et al., 2020] and its 57 subjects, BIG-Bench Hard [Suzgun et al., 2022] and HELM [Liang et al., 2022] are the multi-task suites behind the leaderboards we study; the Open LLM Leaderboard builds on the EleutherAI evaluation harness [Gao et al., 2023]. That the choice of tasks and of how to aggregate them changes who wins has been argued from several sides: the "benchmark lottery" [Dehghani et al., 2021] shows that task selection alters comparisons, Ethayarajh and Jurafsky [2020] critique what leaderboards incentivise, and Shavrina and Malykh [2021] re-rank NLP leaderboards under geometric and harmonic means. Social-choice views of benchmarking apply voting theory to model evaluation: Rofin et al. [2023] propose ordinal aggregation for NLP benchmarks, Lanctot et al. [2023] evaluate agents with social-choice methods, and Zhang and Hardt [2024] use Arrow's theorem to show trade-offs between diversity and stability of multi-task benchmarks. These works ask which rule *should* be used or how *stable* rankings are to the task set; we ask what it *costs* one participant to change the outcome.

Strategic behaviour on leaderboards. Training on test data or on rephrased test data inflates scores [Zhou et al., 2023, Yang et al., 2023], and selective private testing distorts the Chatbot Arena ranking [Singh et al., 2025]; these concern *how* scores are pushed up. The closest work we are aware of is a Stackelberg model in which a benchmark designer chooses the evaluation protocol and developers with latent quality spend resources on benchmark-specific improvement; it shows that current benchmarks induce games without a Nash equilibrium among developers and that testing after tuning restores a unique equilibrium [Chen et al., 2026] (we read its abstract only). That is an equilibrium analysis among many developers with a resource notion of investment; we instead take a single developer's best response, the *minimum* score increase that makes one target first, and measure it on real leaderboards for a family of aggregation rules. More generally, our setting is an instance of optimisation against a proxy metric [Manheim and Garrabrant, 2018].

Manipulation and bribery in voting. Computational social choice studies how hard it is to change an election's outcome by strategic voting, control or bribery [Brandt et al., 2016]; in bribery [Faliszewski et al., 2009] an external actor pays voters to change their preferences, and the question is the

cheapest way to make a chosen candidate win. Our problem is the cardinal-score analogue with the roles reversed (tasks are voters, models are candidates, and the target "bribes" its own scores within a headroom cap), and our rank-based rules are the standard Borda, Condorcet and plurality rules from that literature. New here is the empirical quantity: the minimum cost in score points on real leaderboards, with caps for how far targeted training can plausibly move a score.

3 Problem setup and exact solvers

3.1 The minimum-investment problem

A leaderboard is a matrix $X \in [0, 100]^{M \times n}$ of scores of M models on n tasks (sub-datasets). An *aggregation rule* maps X to a ranking. Fix a *target* model i with score vector $x = X_{i\cdot}$. Targeted investment lets the target move to any y with $x_t \le y_t \le c_t$ for every task t, where $c_t \ge x_t$ is a *headroom cap*; all other rows of X are held fixed. The target is ***first*** if it is strictly ahead of every other model under the rule; for score-based rules with aggregate f this means $f(y) \ge \tau + \varepsilon$ with $\tau = \max_{j \ne i} f(X_{j\cdot})$ and margin $\varepsilon = 10^{-3}$ points, and for rank-based rules strict dominance with ties counted as one half. The *minimum investment* is the least-cost y that makes the target first, or ∞ if no feasible y exists (the target "cannot be gamed" under that cap). We report three costs of a raise $\delta = y - x$:

- $L_1 = \sum_t \delta_t$, total score points (effort, if one point on any task costs the same);
- $K = |\{t : \delta_t > 0\}|$, the number of tasks touched (breadth, i.e. how many sub-datasets' data must be acquired);
- $L_\infty = \max_t \delta_t$, the largest single-task jump; it equals the *uniform* improvement needed if effort cannot be targeted, and so measures the benefit of targeting.

Each measure is minimised separately (so the three numbers need not come from the same raise).

3.2 Headroom caps (the main modelling assumption)

That targeted training can lift a score to *any* value is unrealistic, and the answer to our question depends on how far it can. We therefore treat the cap as the central assumption and vary it: `ceil`: $c_t = 100$ (no restriction beyond the score range); `front0`/`front2`: c_t is the best *other* model's score on t plus 0 or 2 points (training on related data brings the target to, at most, the frontier); `gain5`/`10`/`20`: at most $+5$, $+10$ or $+20$ points per task; `emp95`/`emp99`: the 95th/99th percentile of the fine-tune-minus-base score differences observed on the Open LLM Leaderboard (540 base/fine-tune pairs, per task), an observational calibration of what fine-tuning has achieved, not a causal estimate.

3.3 Aggregation rules

Score-based rules aggregate the score vector: arithmetic **mean**; **geometric** and **harmonic** mean (applied to score$+1$, concave power means with $p \le 0$, as proposed for rate-like metrics [Shavrina and Malykh, 2021]); **median**; **min** (worst-task score); and a **trimmed mean** dropping the best and worst task. *Rank-based rules* use only within-task orderings: **Borda** (mean over tasks of the model's rank score, equivalent to the mean rank); **Condorcet winner** (the target must beat each rival on a strict majority of tasks; it is then first under every Condorcet-consistent rule, e.g. Copeland, Schulze, Kemeny); and **plurality** (number of tasks on which the model is best, ties split). These span the cardinal–ordinal spectrum discussed for benchmarks [Zhang and Hardt, 2024, Rofin et al., 2023].

3.4 Solvers

For the mean, the power means, the median and the min rule the minimum is available in closed form or by water-filling (Section 4). For the three rank-based rules the problem is a mixed-integer program (binary indicators for whether the target beats each rival on each task, with lazily generated rival constraints), solved exactly with HiGHS; L_∞ for rank rules is found by bisection on the feasible uniform-cap level. Every solver is checked against brute-force enumeration on small random instances (7 unit tests in the code repository), and every infinite result from the rank-based rules was re-solved

with a strict mode that raises on a solver time-out: all are proven infeasible rather than unresolved. Table 2 (Appendix B) lists for every rule the condition for first place, the exact solution and the numerical check against the solver; the trimmed mean (dropping the best and worst task) is solved by a small MILP and was run on a sample of Open LLM Leaderboard targets only.

4 Closed forms

Let n be the number of tasks, $H_t = c_t - x_t \geq 0$ the headroom on task t, and assume the target is not already first. The following hold for any leaderboard; Section 5 checks them numerically against the solvers on every target.

Proposition 1 (Mean)**.** *Let $g = \max_{j \neq i} \bar{X}_j - \bar{x}$ be the gap in means. Then the minimum L_1 is $n(g+\varepsilon)$ if $\sum_t H_t \geq n(g+\varepsilon)$ and ∞ otherwise; the minimum K is the least k such that the k largest headrooms sum to at least $n(g+\varepsilon)$; and the minimum L_∞ is the least h with $\sum_t \min(h, H_t) \geq n(g+\varepsilon)$, which is at least $g + \varepsilon$ with equality iff the uniform raise by $g + \varepsilon$ violates no cap.*

Proof. The mean constraint is $\sum_t \delta_t \geq n(g+\varepsilon)$ with $0 \leq \delta_t \leq H_t$; every statement is the elementary solution of this knapsack-type system (take the largest capacities first for K; fill every task to a common level for L_∞). □

Thus under the mean the cost in points depends on the gap only, and the cap matters only for feasibility, for K, and for L_∞: with ample headroom the whole investment can be concentrated on one or two tasks, however many tasks there are.

Proposition 2 (Min)**.** *Let $m = \max_{j \neq i} \min_t X_{jt}$. The target can become first iff $c_t \geq m + \varepsilon$ for all t; then $L_1 = \sum_t (m + \varepsilon - x_t)^+$, $K = |\{t : x_t < m + \varepsilon\}|$ and $L_\infty = \max_t (m + \varepsilon - x_t)^+$.*

Under the min rule investment is forced to be *broad* on every below-threshold task, but it is not scaled by the number of tasks.

Proposition 3 (Median, odd n)**.** *Let $m = \max_{j \neq i} \mathrm{med}(X_{j\cdot})$ and $\mathcal{T} = \{t : c_t \geq m + \varepsilon\}$. The target becomes first iff $|\mathcal{T}| \geq (n+1)/2$; the minimum L_1 is the sum of the $(n+1)/2$ smallest values of $(m + \varepsilon - x_t)^+$ over $t \in \mathcal{T}$, and the minimum K is $(n+1)/2 - |\{t : x_t \geq m + \varepsilon\}|$ (if positive).*

Proof. The median is at least $m+\varepsilon$ iff at least $(n+1)/2$ coordinates are $\geq m+\varepsilon$; raising a coordinate beyond $m + \varepsilon$ is wasted, so the optimum raises the cheapest coordinates exactly to $m + \varepsilon$. □

The median rule needs a *majority* of tasks to be lifted past the strongest rival's median, but only to that level, so its cost is a sum over the cheapest majority and grows faster than linearly in the gap when the target is far from the leader.

Proposition 4 (Concave power means)**.** *For $p \leq 0$ (geometric $p = 0$, harmonic $p = -1$) the rule is $f(y) = \varphi^{-1}(\frac{1}{n}\sum_t \varphi(y_t + 1)) - 1$ with φ increasing and concave. The minimum L_1 equalises $y_t + 1$ at a water level λ clipped to $[x_t, c_t]$ (found by bisection), and the minimum K is a greedy count over the gains $\varphi(c_t + 1) - \varphi(x_t + 1)$.*

Concave means reward balanced improvement and so, compared with the arithmetic mean, favour spreading the investment over low-scoring tasks.

For the rank-based rules we have no closed form; the cost depends on the *pairwise* structure of the leaderboard (how many rivals must be beaten on how many tasks) rather than on a gap in a single aggregate, and is obtained by the mixed-integer solver. One structural fact holds for every rule and is checked numerically: the minimum investment is non-increasing in the caps (a larger headroom set can only help).

5 Experiments

5.1 Data and protocol

Data. *HELM MMLU*: release v1.13.0 of the public HELM MMLU benchmark output (the version fixed by the task), 79 models with a score on all 57 MMLU subjects ($M = 79$, $n = 57$). *Open LLM*

Leaderboard: the public `contents` data set at commit `9c09a7ca` (snapshot of 2025-03-20; parquet SHA-256 `f13a7061...`), 4 576 rows, of which 79 duplicate model names were dropped and one flagged model excluded as a target, leaving 4 496 targets and rivals on $n = 6$ tasks (IFEval, BBH, MATH Lvl 5, GPQA, MuSR, MMLU-Pro), using the leaderboard's *normalised* scores in $[0, 100]$; raw scores are a robustness check. *Granularity board*: the Open LLM Leaderboard `results` data set (commit `aa81ecc3`; newest complete run per model) gives the *same* models on 39 leaf tasks (raw accuracy, 4 506 models), used only in Section 5.6. Neither source declares a licence for the scores; we use the public scores only and redistribute derived score matrices (details and checksums in `REPRODUCE.md`).

Protocol. For every target we compute the minimum investment (Section 3) under the stated rules and caps. Targets are grouped by their rank under the *mean* (the same cohort for every rule): 2, 3–5, 6–10, 11–30, 31–100 and > 100 (Open LLM Leaderboard only); a rank-1 target needs no investment. We report the group median of L_1 and K and the share of targets for which first place is unreachable under the cap; a median is reported as ∞ when more than half of the group is infeasible, and intervals are 1 000-resample bootstrap percentile intervals over targets (seed 0). To compare the two data sets, which have $n = 57$ and $n = 6$ tasks, we plot L_1/n, the mean points per task, because $L_1 = n(g + \varepsilon)$ for the mean. Runs for the rank-based rules on the Open LLM Leaderboard use rank-stratified samples of targets (all ranks 2–21 plus 150 log-uniformly drawn ranks) and are only reported per rank group.

5.2 The closed forms hold

On all 78 non-leading HELM targets and all 4 495 non-leading Open LLM Leaderboard targets the solver's mean-rule L_1 equals $n(g + \varepsilon)$ up to $2.5 \cdot 10^{-12}$ (Proposition 1); $L_\infty \geq g + \varepsilon$, with equality for 4 346 of 4 495 Open LLM and 3 of 78 HELM targets (those whose uniform raise hits no cap); the greedy K matches with no mismatch; the min rule matches Proposition 2 on all targets, and the median's K matches Proposition 3 on HELM ($n = 57$ is odd; $n = 6$ is even and solved exactly but not covered). Costs are monotone in the caps along the `ceil`/`front` and `gain` chains for every rule (`results/theory_checks.txt`). These identities are consequences of the model, not discoveries; they validate the solvers.

5.3 Cost versus rank and rule

Figure 1 and Table 1 give the headline picture for the cap that is most favourable to the developer (`ceil`, 100 points); a bounded gain (`gain10`) is in Figure 2, Table 1 and Appendix Figure 7.

Under the mean the cost grows with rank at n times the gap and is the highest of all rules. On HELM with `ceil`, making a target ranked 11–30 first costs a median of 435 points under the mean (7.6 per task, $K = 10$ tasks) and 1 021 (17.9 per task, $K = 21$) at ranks 31–100; the geometric and harmonic means are within 6–10% of it. On the Open LLM Leaderboard the mean costs 32 points (5.4 per task) at ranks 11–30 and 102 (17 per task) beyond rank 100, with K of 1 or 2 tasks: one task suffices because the headroom is large. **Median, min and plurality are much cheaper than the mean on HELM.** At ranks 11–30 the median costs 38 points, min 46 and plurality 25 (mean 435); at ranks 31–100, 207, 200 and 72 (mean 1 021). The Condorcet requirement costs 107 and 312, Borda 286 and 858. **The cost is differently shaped, not just smaller.** The Condorcet requirement forces breadth ($K = 28$–29 of 57 tasks) at small L_1, Borda needs up to 51 tasks, whereas the mean-like rules and min need 7–21 (Table 1): a developer who can acquire data for few sub-datasets faces a different ranking of rules than one who counts points. **The ordering does not carry over to a different board.** On the Open LLM Leaderboard ($n = 6$) the median is the cheapest rule at ranks 6–10 and 11–30 (0.06 and 0.71 points per task versus 4.6 and 5.4 for the mean) but costs more than the mean at rank 2 (1.59 versus 0.79, one target); Borda, among the most expensive rules on HELM, is one of the cheapest under bounded gains (1.57 points per task at ranks 11–30 under `gain10` versus 5.36 for the mean). The cost of the rank-based rules depends on the board, so no statement of the form "rank rules are easier (or harder) to game" is supported; Section 5.6 traces the dependence to the number of tasks. **Rules agree on which targets are cheap.** Across HELM targets the Spearman correlation of L_1 among mean, geometric, harmonic, Borda, Condorcet and median is 0.92–1.00 under `gain10` (plurality 0.87–0.98; min is the outlier, 0.67–0.78; Appendix Table 4).

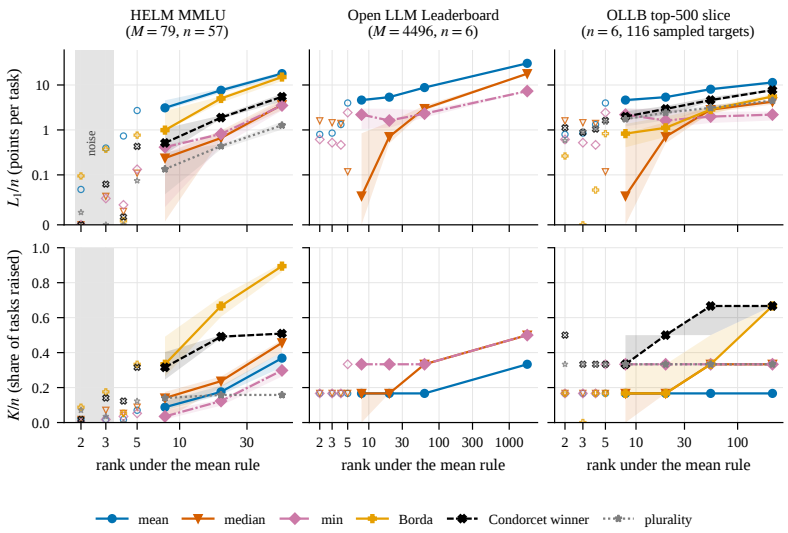


Figure 1: Other rules make first place much cheaper in points than the mean, but not in tasks: on HELM the median, min and plurality need a fraction of the mean's points per task, whereas Borda and the Condorcet requirement need to raise many tasks. Minimal investment to rank first (cap: scores may rise to 100) by the target's rank under the mean rule; top: points raised per task L_1/n (symmetric log axis, 0 = already first), bottom: share of tasks raised K/n. Lines are medians over the targets of a rank group (6–10, 11–30, 31–100, >100; HELM has no targets beyond rank 79, so its last group is 31–78), bands are 95% bootstrap CIs over targets (1000 resamples, seed 0); hollow markers are individual targets in the groups with fewer than five targets (ranks 2 and 3–5). In the shaded band, HELM's rank-2 and rank-3 targets trail the leader by only 0.14 and 0.8 standard errors of the gap (Table 9), so their costs are noise-limited. Left to right: HELM MMLU (78 targets), OLLB (4495 targets; ordinal rules not run on the full board) and the top-500 OLLB models (116 rank-stratified targets, all rules). The Condorcet winner is an upper bound for Copeland and Schulze.

5.4 The cap decides whether first place is reachable

Figure 2 shows what happens when the headroom is bounded.

With `ceil` every target can be made first under every rule that was run with it (no infeasible group on either board), but with `gain10` on HELM *no* target ranked 31–100 can reach first under the mean, Borda, geometric or harmonic rule (100% infeasible; Condorcet 82%, median 78%, plurality 43%), and at ranks 11–30 the min rule is already infeasible for 70% of targets. On the Open LLM Leaderboard under `gain10` 96% of the targets ranked worse than 100 cannot be made first by the mean, and under `gain5` most targets ranked 11–30 cannot either (80% under the mean). The data-grounded cap `emp95` gives the same picture for the mean rule: 95% of the targets ranked worse than 100 are out of reach. These infeasible shares are as informative as the costs: with a bounded per-task gain, *whether a rule can be gamed at all* depends on the target's distance to the leader. Every headline statement above is therefore about a cap: with `ceil` the costs are lower bounds on what any developer would pay.

5.5 Scaling with the gap and with the number of tasks

On HELM under `ceil`, the log–log slope of L_1 on the target's own gap is 1.00 for the mean (Proposition 1), 1.05 [1.04, 1.06] for the geometric and 1.10 [1.07, 1.11] for the harmonic mean, but 1.63 [1.57, 1.67] for the median and 1.69 [1.54, 1.92] for the min: order-statistic rules are cheap only for targets already close to the leader, and their cost grows super-linearly with distance (Open LLM Leaderboard: 1.26, 1.48, 1.20, 1.65). Over random subsets of $n = 3, \ldots, 48$ HELM subjects (200 draws; rank-5 target, `ceil`), the mean rises from 7.0 points at $n = 3$ to 129 at $n = 48$ (geometric 6.8 to 117, Borda 3.3 to 42), whereas median and min stay between 1.9 and 7.2 points for every n and plurality between 2.3 and 3.0 for $n \geq 6$ (20.8 at $n = 3$) (Figure 3c, Appendix Figure 4): the more tasks, the cheaper these rules are relative to the mean.

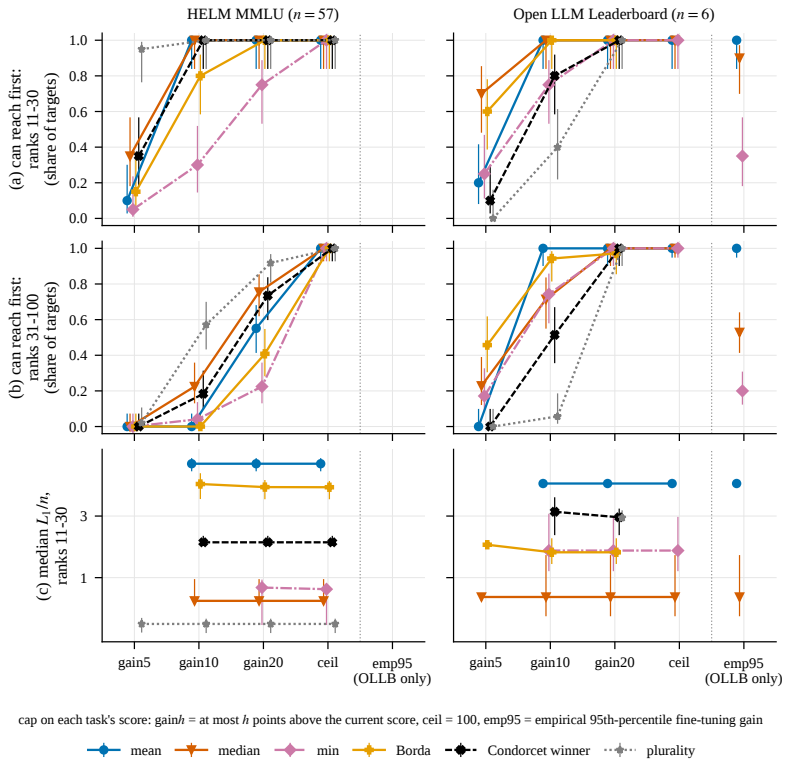


Figure 2: Whether first place is reachable at all depends on the cap on per-task gains and on the rule, so no cost statement holds without its cap. (a, b) Share of targets that can reach first place under each cap, for targets ranked 11–30 (a) and 31–100 (b) under the mean rule; bars are 95% Wilson intervals. (c) Median L_1/n of the 11–30 cohort; no point where more than half of the cohort is infeasible; bars are 95% bootstrap CIs over targets. Caps: gain$h$ = at most h points above the current score on each task, ceil = 100, emp95 = empirical 95th-percentile fine-tuning gain per task (OLLB only). Left: HELM (20 and 49 targets in the two cohorts). Right: OLLB, where all rules are compared on the targets common to all rules run under a cap (Borda, Condorcet and plurality were run only for gain caps, on a rank-stratified sample: 20 and 35 targets in the two cohorts); missing symbols mean not run. The Condorcet winner is an upper bound for Copeland and Schulze. Bootstrap resamples 1000 (seed 0); input hashes in `figures/inputs.sha256`.

5.6 What drives the differences between rules

The two boards differ in both M and n, so the cross-board differences above confound the two. **Number of tasks, not models.** The Open LLM Leaderboard `results` data set gives the same 4 506 models on 39 leaf tasks and, aggregated item-weighted, on six benchmarks (raw accuracy; rank groups under each board's own mean; `ceil`). The mean costs about the same per task on both (rank > 100: 25.3 versus 26.0 points; at ranks 11–30 the cohorts differ, 3.0 versus 4.9), as $L_1/n = g + \varepsilon$ requires. The median and min rules become 5–20 times cheaper per task on the finer board at ranks 11–100 (2–3 times beyond rank 100): at ranks 11–30 the median costs 0.22 versus 4.44 points per task and min 0.27 versus 2.40; at ranks 31–100, 0.61 versus 5.44 and 0.47 versus 2.59; the median needs 9% of the tasks instead of 33% (Figure 3; Appendix Tables 5, 6). Under `gain10` they become reachable where the six-task board is not (rank 11–30 median: 0% versus 80% infeasible), whereas plurality becomes unreachable (80% versus 10% of targets at ranks 11–30 on the top-500 slices), since many first places are needed. Random task subsets of the 39-task board agree (rank-5 target, `ceil`, median points per task at $n = 6/24/39$: mean 3.25/1.56/1.64, median 1.19/0.20/0.055, min 0.54/0.080/0.050): the median-to-mean cost ratio falls from 0.36 to 0.034. The number of tasks drives the divergence between order-statistic rules and the mean, as on HELM; caveat: leaf scores are raw accuracies (Section 5.7). **Targeting is what makes ordinal rules cheap.** If the investment is spread over all tasks at random instead of optimised (HELM, targets ranked 2–20, 100 draws per target,

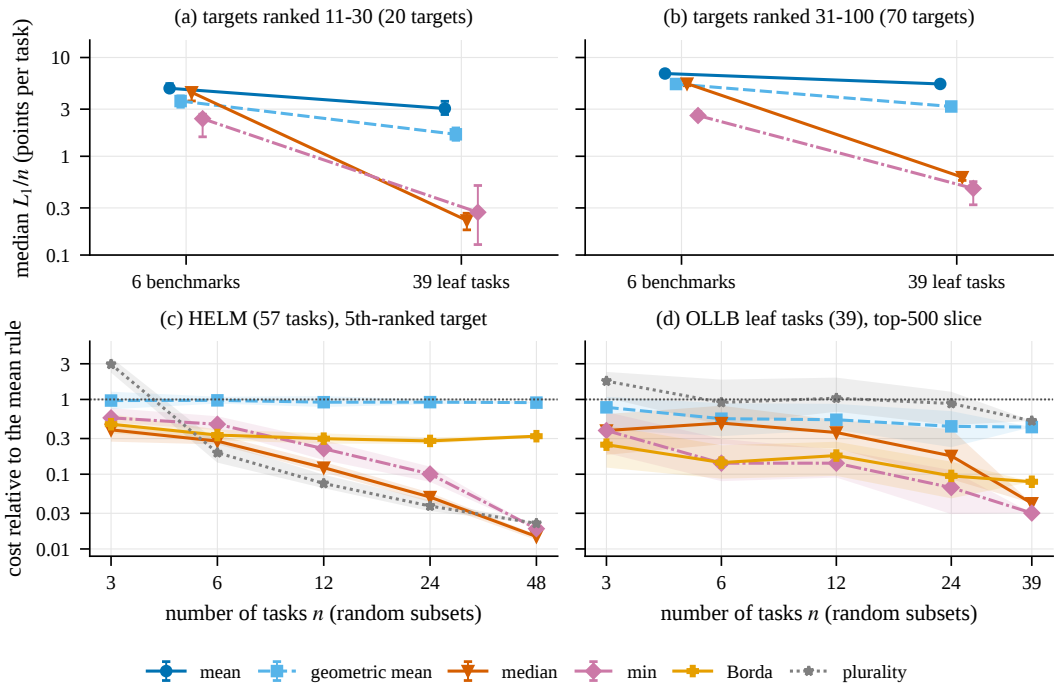


Figure 3: The number of tasks n, not the number of models, drives how much cheaper the median and min rules are than the mean: their cost relative to the mean falls from about 0.4 at $n = 3$ to 0.015–0.04 on the full boards, while the mean's own cost per task does not change. (a, b) The same 4506 Open LLM Leaderboard models scored on six benchmark aggregates and on 39 leaf tasks (raw accuracy): median L_1/n (ceil cap) of the targets ranked 11–30 (a) and 31–100 (b) under each board's own mean rule, 20 and 70 sampled targets per board; bars are 95% bootstrap CIs over targets. (c, d) Random task subsets, ceil cap, target ranked 5th under each rule's own ranking: median L_1/n of a rule divided by the mean rule's, for HELM (c, $n \leq 48$ of 57 tasks, 200 draws) and for the top-500 slice of the leaf board (d, 60 draws; $n = 39$ is the full board, one subset). Bands are the widest ratio the two 95% draw-CIs allow (not a CI of the ratio). Leaf scores are raw accuracies. Absolute costs: Table 5, Figure 4; gain10: Figure 6.

`ceil`), the budget has to be multiplied by 1.00 for the mean (any split works while no cap binds), 1.2 and 1.4 for the geometric and harmonic means, 2.9 for Borda, 4.0 for the Condorcet requirement, 7.7 for plurality, 12 for the median and 38 for min (median over targets; `gain10`: 1.0, 1.1, 1.2, 2.4, 3.7, 7.7, 12, 43; Appendix Table 7). The rules that are cheapest with optimal targeting are exactly those that profit most from it. Raising a random set of as many tasks as the optimum uses (all the way to the cap) fails for 90–100% of the targets under every rule, including the mean: which tasks are raised matters, not only how many. **The Condorcet requirement is a bound.** An exact Copeland solver (34 target–cap cases) gives, on HELM (ranks 2–70), a cost equal to the Condorcet-winner cost minus one margin in every feasible case: becoming a Condorcet winner is the cheapest way to top Copeland there. On the top-500 Open LLM slice Copeland is 1.4–8.1 times cheaper than the Condorcet requirement and always cheaper than the mean rule, so there the requirement is only an upper bound (Appendix Table 8). Schulze and Kemeny were not solved.

5.7 Robustness

Normalisation and slices. On raw Open LLM scores the advantage of the median over the mean shrinks and min becomes costlier (ranks 11–30, `ceil`: median 20.8 versus mean 23.0 points, against 4.2 versus 32.2 normalised). Removing merged models changes group medians by at most 20% for the mean and up to 29% for min in small rank groups, with the order of the score rules unchanged; the top-500 slice leaves the score rules' order unchanged but moves the rank-based rules. **Evaluation noise and task choice.** Redrawing HELM scores from their binomial standard errors, or resampling tasks with replacement, 100 times for the ten best targets under `gain10`: rules cheap on the observed

data stay cheap (share of draws cheaper than the mean: plurality 0.94, median 0.93–0.95, Condorcet 0.94–0.95, Borda 0.90–0.92), but the min rule's cost is unstable (within-target coefficient of variation 1.2–1.3, 30–44% of draws infeasible) and the median's noisy (0.58–0.68); the mean and geometric mean are most stable (0.17–0.26). The observed cost of a rule thus overstates how reliably it can be gamed under noisy scores (Appendix Figure 8).

6 Discussion and limitations

What the results say, and for whom. The investment needed to make a target first is determined jointly by the rule, the headroom cap, and the number of tasks and models of the board. Under the arithmetic mean, the Open LLM Leaderboard's own rule, the cost in points is exactly n times the gap and can be concentrated on one or two tasks whenever there is headroom; under a bounded 10-point gain, most mid- and low-ranked targets cannot be made first at all. For a leaderboard designer the reading is not "prefer rule X": the same rules that are cheap to game on a 79-model, 57-task board (median, min, plurality) are less so, or differently so, on a 4 496-model, 6-task board. The granularity experiment (Section 5.6) attributes this to the number of tasks: the same models on 39 leaf tasks make the median and min rules 5–20 times cheaper per task at ranks 11–100 while the mean is unchanged, so in our data adding finer tasks to a board aggregated by an order statistic lowers the cost of topping it, and under bounded gains raises it for plurality. We did not measure the cost of obtaining a given gain.

Modelling assumptions and where they fail. (i) *Reachability.* A score can be raised to any value up to a cap and raising one task leaves the others unchanged; real fine-tuning can lower other tasks, and the caps are the only handle on attainability. The `emp95/emp99` caps are calibrated on 540 self-selected base/fine-tune pairs, not a causal estimate of what training does. (ii) *Cost.* Investment is measured in score points and tasks touched, not in data, compute or money; comparing rules by L_1 assumes a point costs the same on every task, which is false in general (hence K and L_∞). (iii) *Static rivals, one target.* Other models do not react, one developer acts, and the target's own score noise is not modelled in the main runs (only in the Monte-Carlo runs); a game among developers is a different question [Chen et al., 2026]. (iv) *Margin.* "First" means ahead by 10^{-3} points; costs of about 0.002 reflect this margin.

Limitations of the evidence. HELM has 79 models, so rank groups 2, 3–5 and 6–10 hold 1, 3 and 5 targets and their intervals are wide or degenerate; rank dependence is read from the Open LLM Leaderboard ($n = 6$, so $K \leq 6$). The rank-based rules there were run on rank-stratified samples of targets with gain caps (and on the top-500 slice for all caps), not on all targets with `ceil`/`front` caps, so only per-group medians are meaningful. The Condorcet-winner requirement is exact for Copeland on HELM but an upper bound on the top-500 Open LLM slice, where Copeland is up to 8 times cheaper (Section 5.6); Schulze and Kemeny were not solved. On HELM the leader is ahead of the second model by 0.07 points, 0.14 binomial standard errors of the gap (an upper bound, since models share items); ranks 3 and 4 are 0.8 and 1.5 standard errors behind (ranks 5–8: 5.4–6.2), so the rank-2 to rank-4 investments lie inside evaluation noise (Appendix Table 9) and we do not read them as a ranking of rules. Results are for one snapshot of each board; the Open LLM Leaderboard contains merged models (708 of 4 497) and near-duplicates, and removing them changes group medians by at most 20% (mean) and 29% (min) in small rank groups. The granularity board uses raw accuracies. A finding that a rule is cheap to game concerns the scores, not any model (the near-tie table names targets only to identify them); we recommend no action.

Broader impact. The analysis could in principle be used to plan targeted training; the data are public, the model is deliberately coarse (no estimate of what any lab could achieve), and we expect the main use in leaderboard design.

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A Proof of Proposition 4 and notes on the other closed forms

Concave power means. Write $z_t = y_t + 1$, $\varphi(z) = \log z$ for $p = 0$ and $\varphi(z) = (z^p - 1)/p$ for $p < 0$; φ is increasing and strictly concave and $f(y) \geq \theta$ iff $\sum_t \varphi(y_t + 1) \geq n\varphi(\theta + 1) =: B$ (with $\theta = \tau + \varepsilon$). Minimise $\sum_t (y_t - x_t)$ over $x_t \leq y_t \leq c_t$ subject to $\sum_t \varphi(y_t + 1) \geq B$. The problem is convex (linear objective, concave constraint), so the KKT conditions are sufficient: there is $\mu \geq 0$ with $\varphi'(y_t + 1) = 1/\mu$ on tasks with $x_t < y_t < c_t$, $\varphi'(y_t + 1) \leq 1/\mu$ where $y_t = x_t$

is binding below, and $\varphi'(y_t + 1) \geq 1/\mu$ where $y_t = c_t$. Because φ' is strictly decreasing this is $y_t = \min\{c_t, \max\{x_t, \lambda\}\}$ with common level $\lambda = 1/\varphi'^{-1}(\cdot) - 1$, i.e. water-filling. The left-hand side $\sum_t \varphi(\min\{c_t, \max\{x_t, \lambda\}\} + 1)$ is continuous and non-decreasing in λ, so the level is found by bisection (feasible iff it reaches B at $\lambda = \max_t c_t$). For K, the constraint is additive in the gains $\varphi(y_t + 1) - \varphi(x_t + 1) \geq 0$, each task contributes at most $\varphi(c_t + 1) - \varphi(x_t + 1)$ when raised, and raising a task to its cap is optimal when it is raised at all, so taking the tasks in decreasing order of that gain until B is reached minimises the number of tasks. □

Mean (Proposition 1), L_∞. For a common raise level h the largest total gain attainable is $\sum_t \min(h, H_t)$, which is continuous and increasing in h; the smallest h with $\sum_t \min(h, H_t) \geq n(g + \varepsilon)$ is the minimum L_∞, and $h \geq g + \varepsilon$ because $\sum_t \min(h, H_t) \leq nh$.

Median, even n. For even n the median is the mean of the two middle order statistics; the solver enumerates the pair of thresholds and solves a sorted-cost selection for each (exact, but without a short closed form), and the statement of Proposition 3 is restricted to odd n.

B Additional figures and tables

Table 1: The cheapest rule depends on the cap and the board, and under a bound of 10 points per task most rules cannot lift a low-ranked HELM target to first place. Minimal investment to rank first, by rule, for targets ranked 11–30 and 31–100 under the mean rule (1000 bootstrap resamples, seed 0). Cells are medians over the targets with the 95% bootstrap CI over targets in brackets; L_1/n = points per task, K = tasks raised. (x% inf.) = share of the cohort that cannot reach first under the cap; ∞ where more than half can not. N = targets in the two cohorts. OLLB: Borda, Condorcet and plurality were not run on the full board at the ceil cap. Ranks 2–10 are omitted because they contain only 1, 3 and 5 HELM targets. Condorcet winner (a sufficient but not necessary condition for first under Copeland and Schulze) is an upper bound for those rules.

	ranks 11–30		ranks 31–100	
rule	L_1/n	K	L_1/n	K
HELM MMLU, ceil cap (n=57; N = 20 / 49 targets)				
mean	7.6 [6.7, 8.4]	10 [9, 12]	18 [15, 20]	21 [18, 23]
median	0.66 [0.63, 0.97]	14 [12, 15]	3.6 [2.8, 4.8]	26 [23, 27]
min	0.81 [0.43, 0.88]	7 [5.5, 7.5]	3.5 [2.7, 4.3]	17 [15, 18]
Borda	5 [4, 5.6]	38 [34, 41]	15 [12, 17]	51 [49, 52]
Condorcet winner	1.9 [1.8, 2.1]	28 [28, 28]	5.5 [4.3, 6.4]	29 [29, 29]
plurality	0.44 [0.37, 0.48]	9 [9, 9]	1.3 [1, 1.5]	9 [9, 9]
HELM MMLU, gain10 cap (n=57; N = 20 / 49 targets)				
mean	7.6 [6.7, 8.4]	44 [39, 49]	∞ (100% inf.)	∞
median	0.66 [0.63, 0.97]	14 [12, 15]	∞ (78% inf.)	∞
min	∞ (70% inf.)	∞	∞ (96% inf.)	∞
Borda	5.3 [4.1, 6.4] (20% inf.)	50 [44, 54]	∞ (100% inf.)	∞
Condorcet winner	1.9 [1.8, 2.1]	28 [28, 28]	∞ (82% inf.)	∞
plurality	0.44 [0.37, 0.48]	9 [9, 9]	1.3 [1, ∞] (43% inf.)	10 [10, ∞]
OLLB, ceil cap (n=6; N = 20 / 70 targets)				
mean	5.4 [5.3, 5.6]	1 [1, 1]	8.8 [8.5, 9]	1 [1, 1]
median	0.71 [0.5, 1.5]	1 [1, 1]	3 [2.7, 3.5]	2 [2, 2]
min	1.6 [1.1, 3]	2 [2, 2]	2.3 [1.9, 2.7]	2 [2, 2]

Table 3: Minimal investment by rule and cap for the cohorts of Table 2 (cap scenarios not shown there; geometric and harmonic means added). Notation and conventions as in Table 2; OLLB gain and emp caps compare all rules on the targets common to all rules run under the cap.

	ranks 11–30		ranks 31–100	
rule	L_1/n	K	L_1/n	K
HELM MMLU, front0 cap (n=57; N = 20 / 49 targets)				
mean	7.6 [6.7, 8.4]	28 [26, 29]	18 [15, 20]	42 [38, 44]

	ranks 11–30		ranks 31–100	
rule	L_1/n	K	L_1/n	K
median	0.7 [0.66, 1]	14 [12, 15]	3.9 [3.1, 5.1]	26 [23, 27]
min	0.81 [0.43, 0.88]	7 [5.5, 7.5]	3.5 [2.7, 4.3]	17 [15, 18]
Borda	5.3 [4.3, 5.9]	40 [36, 43]	15 [13, 17]	52 [50, 53]
Condorcet winner	2.2 [2.1, 2.6]	28 [28, 29]	7 [5.8, 7.9]	32 [32, 32]
plurality	1.5 [1.3, 1.6]	18 [18, 18]	3.7 [3, 4.4]	18 [18, 18]
geometric mean	7.4 [6.4, 8.2]	27 [24, 28]	18 [15, 19]	41 [37, 43]
harmonic mean	7.2 [6.2, 8]	26 [24, 27]	17 [15, 19]	40 [37, 43]
HELM MMLU, front2 cap (n=57; N = 20 / 49 targets)				
mean	7.6 [6.7, 8.4]	22 [21, 24]	18 [15, 20]	36 [33, 38]
median	0.66 [0.63, 0.97]	14 [12, 15]	3.6 [2.8, 4.8]	26 [23, 27]
min	0.81 [0.43, 0.88]	7 [5.5, 7.5]	3.5 [2.7, 4.3]	17 [15, 18]
Borda	5 [4, 5.6]	38 [34, 41]	15 [12, 17]	51 [49, 52]
Condorcet winner	1.9 [1.8, 2.1]	28 [28, 28]	5.5 [4.3, 6.4]	29 [29, 29]
plurality	0.44 [0.37, 0.48]	9 [9, 9]	1.3 [1, 1.5]	9 [9, 9]
geometric mean	7.3 [6.4, 8.1]	21 [20, 23]	18 [15, 19]	35 [32, 38]
harmonic mean	7.1 [6.1, 7.8]	20 [20, 22]	17 [14, 19]	35 [32, 38]
HELM MMLU, gain5 cap (n=57; N = 20 / 49 targets)				
mean	∞ (90% inf.)	∞	∞ (100% inf.)	∞
median	∞ (65% inf.)	∞	∞ (100% inf.)	∞
min	∞ (95% inf.)	∞	∞ (100% inf.)	∞
Borda	∞ (85% inf.)	∞	∞ (100% inf.)	∞
Condorcet winner	∞ (65% inf.)	∞	∞ (100% inf.)	∞
plurality	0.44 [0.38, 0.49] (5% inf.)	10 [9, 10]	∞ (98% inf.)	∞
geometric mean	∞ (100% inf.)	∞	∞ (100% inf.)	∞
harmonic mean	∞ (100% inf.)	∞	∞ (100% inf.)	∞
HELM MMLU, gain20 cap (n=57; N = 20 / 49 targets)				
mean	7.6 [6.7, 8.4]	22 [20, 24]	18 [15, ∞] (45% inf.)	57 [45, ∞]
median	0.66 [0.63, 0.97]	14 [12, 15]	3.6 [2.8, 4.8] (24% inf.)	26 [23, 27]
min	0.84 [0.43, 0.97] (25% inf.)	7 [5.5, 9]	∞ (78% inf.)	∞
Borda	5 [4, 5.7]	40 [38, 44]	∞ (59% inf.)	∞
Condorcet winner	1.9 [1.8, 2.1]	28 [28, 28]	5.5 [4.3, 6.5] (27% inf.)	29 [29, 29]
plurality	0.44 [0.37, 0.48]	9 [9, 9]	1.3 [1, 1.5] (8% inf.)	9 [9, 9]
geometric mean	7.4 [6.4, 8.3]	22 [19, 25]	∞ (57% inf.)	∞
harmonic mean	7.2 [6.1, 8.1]	21 [18, 24]	∞ (57% inf.)	∞
OLLB, front0 cap (n=6; N = 20 / 70 targets)				
mean	5.4 [5.3, 5.6]	2 [2, 2]	8.8 [8.5, 9]	2 [2, 3]
median	0.71 [0.5, 1.5]	1 [1, 1]	3 [2.7, 3.5]	2 [2, 2]
min	1.6 [1.1, 3]	2 [2, 2]	2.3 [1.9, 2.7]	2 [2, 2]
geometric mean	3.9 [3.6, 4.4]	2 [2, 2]	6.3 [6, 6.5]	2 [2, 2]
harmonic mean	3 [2.5, 4]	2 [1.5, 2]	4.4 [4, 4.8]	2 [2, 2]
OLLB, front2 cap (n=6; N = 20 / 70 targets)				
mean	5.4 [5.3, 5.6]	2 [2, 2]	8.8 [8.5, 9]	2 [2, 2]
median	0.71 [0.5, 1.5]	1 [1, 1]	3 [2.7, 3.5]	2 [2, 2]
min	1.6 [1.1, 3]	2 [2, 2]	2.3 [1.9, 2.7]	2 [2, 2]
geometric mean	3.9 [3.6, 4.4]	2 [2, 2]	6.2 [5.7, 6.4]	2 [2, 2]
harmonic mean	3 [2.5, 4]	2 [1, 2]	4.3 [4, 4.8]	2 [2, 2]
OLLB, gain5 cap (n=6; N = 20 / 35 targets)				
mean	∞ (80% inf.)	∞	∞ (100% inf.)	∞
median	0.71 [0.5, ∞] (30% inf.)	1 [1, ∞]	∞ (77% inf.)	∞
min	∞ (75% inf.)	∞	∞ (83% inf.)	∞
Borda	1.8 [1.3, ∞] (40% inf.)	3 [2, ∞]	∞ (54% inf.)	∞
Condorcet winner	∞ (90% inf.)	∞	∞ (100% inf.)	∞
plurality	∞ (100% inf.)	∞	∞ (100% inf.)	∞
geometric mean	∞ (90% inf.)	∞	∞ (100% inf.)	∞
harmonic mean	∞ (85% inf.)	∞	∞ (91% inf.)	∞
OLLB, gain10 cap (n=6; N = 20 / 35 targets)				
mean	5.4 [5.3, 5.6]	4 [4, 4]	8 [7.4, 8.5]	5 [5, 6]
median	0.71 [0.5, 1.5]	1 [1, 1.5]	2.8 [2.4, 3.7] (29% inf.)	3 [2, 3]
min	1.6 [1.1, 3.1] (25% inf.)	2 [2, 2]	2 [1.4, 2.8] (26% inf.)	2 [2, 2]
Borda	1.6 [1.3, 2]	2 [1.5, 2]	2.9 [2.3, 3.4] (6% inf.)	3 [3, 3]
Condorcet winner	3.2 [2.1, 4.2] (20% inf.)	4 [3, 4]	5.9 [4.4, ∞] (49% inf.)	5 [5, ∞]

	ranks 11–30		ranks 31–100	
rule	L_1/n	K	L_1/n	K
plurality	∞ (60% inf.)	∞	∞ (94% inf.)	∞
geometric mean	4.4 [3.8, 5.9]	3 [3, 4]	7 [6.2, 8.7] (17% inf.)	5 [4, 6]
harmonic mean	3 [2.5, 6.7] (15% inf.)	2 [2, 4.5]	4.7 [3.7, 7.9] (26% inf.)	3 [3, 5]
OLLB, gain20 cap (n=6; N = 20 / 35 targets)				
mean	5.4 [5.3, 5.6]	2 [2, 2]	8 [7.4, 8.5]	3 [3, 3]
median	0.71 [0.5, 1.5]	1 [1, 1]	2.8 [2.3, 3.4]	2 [2, 2]
min	1.6 [1.1, 3]	2 [2, 2]	2 [1.4, 2.8]	2 [2, 2]
Borda	1.6 [1.3, 2]	1.5 [1, 2]	2.7 [2.3, 3.4] (3% inf.)	2 [2, 3]
Condorcet winner	2.9 [2.1, 3.4]	3 [3, 3]	4.4 [3.9, 5]	4 [3, 4]
plurality	2.9 [2.7, 3.3]	2 [2, 2]	3.9 [3.5, 4.1]	2 [2, 2]
geometric mean	3.9 [3.6, 4.4]	2 [2, 2]	5.4 [5.3, 6]	2 [2, 2]
harmonic mean	3 [2.5, 4]	2 [1.5, 2]	3.8 [3.5, 4.4]	2 [2, 2]
OLLB, emp95 cap (n=6; N = 20 / 70 targets)				
mean	5.4 [5.3, 5.6]	2 [2, 2]	8.8 [8.5, 9]	4 [3, 4]
median	0.71 [0.5, 1.5] (10% inf.)	1 [1, 1]	4.1 [3.7, ∞] (47% inf.)	3 [3, ∞]
min	∞ (65% inf.)	∞	∞ (80% inf.)	∞
geometric mean	6.1 [5.4, ∞] (30% inf.)	4 [3, ∞]	∞ (60% inf.)	∞
harmonic mean	8.4 [5, ∞] (40% inf.)	5.5 [3.5, ∞]	∞ (70% inf.)	∞
OLLB, emp99 cap (n=6; N = 20 / 70 targets)				
mean	5.4 [5.3, 5.6]	2 [2, 2]	8.8 [8.5, 9]	3 [3, 3]
median	0.71 [0.5, 1.5]	1 [1, 1]	3.3 [2.9, 3.6] (1% inf.)	2 [2, 2.5]
min	1.6 [1.1, ∞] (40% inf.)	2 [2, ∞]	∞ (57% inf.)	∞
geometric mean	4.6 [4, 6.4]	3 [2, 3]	8.4 [7.6, 9.4] (4% inf.)	4 [4, 5]
harmonic mean	3 [2.5, 9.4] (20% inf.)	2 [2, 5]	9 [5.9, 11] (36% inf.)	4.5 [3, 6]
OLLB top-500 slice, ceil cap (n=6; N = 20 / 43 targets)				
mean	5.4 [5.3, 5.6]	1 [1, 1]	8 [7.8, 8.7]	1 [1, 1]
median	0.71 [0.5, 1.5]	1 [1, 1]	2.8 [2.4, 3.5]	2 [2, 2]
min	1.6 [1.1, 3]	2 [2, 2]	2 [1.4, 2.6]	2 [2, 2]
Borda	1.1 [0.7, 2]	1 [1, 2]	2.9 [2.2, 3.3]	2 [2, 3]
Condorcet winner	2.9 [2.1, 3.4]	3 [3, 3]	4.6 [4.2, 5.4]	4 [3, 4]
plurality	2.4 [1.6, 2.6]	2 [2, 2]	3.1 [2.8, 3.6]	2 [2, 2]
geometric mean	3.9 [3.6, 4.4]	1 [1, 1]	5.5 [5.3, 6]	1 [1, 1]
harmonic mean	3 [2.5, 4]	1 [1, 1]	4.1 [3.5, 4.5]	1 [1, 1]
OLLB top-500 slice, gain10 cap (n=6; N = 20 / 43 targets)				
mean	5.4 [5.3, 5.6]	4 [4, 4]	8 [7.8, 8.7]	5 [5, 6]
median	0.71 [0.5, 1.5]	1 [1, 1.5]	2.8 [2.4, 3.8] (33% inf.)	3 [2, 3]
min	1.6 [1.1, 3.1] (25% inf.)	2 [2, 2]	2 [1.4, 2.7] (23% inf.)	2 [2, 2]
Borda	1.1 [0.7, 2]	2 [1, 2]	2.9 [2.2, 3.3] (2% inf.)	3 [3, 3]
Condorcet winner	3.2 [2.1, 4.2] (20% inf.)	4 [3, 4]	∞ (51% inf.)	∞
plurality	2.4 [1.6, 2.6] (10% inf.)	2 [2, 2]	∞ (63% inf.)	∞
geometric mean	4.4 [3.8, 5.9]	3 [3, 4]	7.3 [6.2, 8.3] (9% inf.)	5 [4, 5]
harmonic mean	3 [2.5, 6.7] (15% inf.)	2 [2, 4.5]	5.2 [3.7, 8.2] (26% inf.)	4 [3, 5]

C Reproducibility, compute and disclosure

Data and versions. HELM MMLU release v1.13.0 (79 models, 57 subjects) and the Open LLM Leaderboard `contents` data set at commit `9c09a7ca` (parquet SHA-256 `f13a7061...`); the full checksums, the filtering counts and the licence situation are in `REPRODUCE.md`. Neither source declares a licence for its scores.

Compute. All experiments are CPU-only (2 CPUs, 4 GB memory; no job above 2 GB). The full pipeline (tests, all experiments, summaries and numerical checks) takes 3.75 h of wall-clock time back to back, measured on a shared host: the sampled rank-based runs on the Open LLM Leaderboard take 1.6 h, HELM and the other core runs 0.7 h, the robustness runs 0.5 h, the granularity board 0.5 h, the Copeland and random-allocation controls 0.4 h. Every job is single-threaded (about 3–4 CPU-hours in total). Solvers are deterministic; the bootstrap, evaluation-noise, task-bootstrap, task-subset and random-allocation runs use seed 0. Commands and the map from each table and figure to its generating script are in `REPRODUCE.md`.

Table 2: Aggregation rules, the condition for a target to be first, and the minimal-investment solution. x = the target's scores, y = its scores after investment ($x_t \le y_t \le \mathrm{cap}_t$), τ = the best rival's value under the rule, $\varepsilon = 10^{-3}$ points, $b = \tau + \varepsilon$, gap $= \tau - \bar{x}$ for the mean. Last column: outcome of the numerical checks of the closed forms against the solver output on every target (HELM, OLLB).

rule	target is first iff	minimal investment (exact)	checked
mean	$\frac{1}{n}\sum_t y_t \ge \tau + \varepsilon$	$L_1 = n(\tau - \bar{x} + \varepsilon) = n(\mathrm{gap} + \varepsilon)$; K: fewest tasks whose largest headrooms sum to L_1; $L_\infty \ge \mathrm{gap} + \varepsilon$, equality if the uniform raise is uncapped	Prop. 1: HELM pass, OLLB pass
geometric, harmonic	power mean of $(y_t + 1)$ with $p \in \{0, -1\} \ge \tau + \varepsilon$	water-filling: raise the lowest tasks to a common level (bisection); K by greedy on the gain of $g(\mathrm{cap}) - g(x)$	vs. brute force (unit tests)
median	median of $y \ge \tau + \varepsilon$	odd n: $K = \max\{0, \frac{n+1}{2} - \#\{t : x_t \ge b\}\}$, $b = \tau + \varepsilon$; L_1: the $\frac{n+1}{2}$ cheapest raises to b; even n: exact search	Prop. 3 (K, odd n): HELM pass
min	$\min_t y_t \ge \tau + \varepsilon$	$L_1 = \sum_t (b - x_t)_+$, $K = \#\{t : x_t < b\}$, $L_\infty = b - \min_t x_t$; infeasible iff some cap $< b$	Prop. 2: HELM pass, OLLB pass
Borda	most Borda points (= lowest mean rank)	exact MILP, lazily generated rival constraints; no closed form	vs. brute force (unit tests)
Condorcet winner	beats every rival on a strict majority of tasks	exact MILP; cost of first under every Condorcet-consistent rule (an upper bound for any single such rule)	vs. brute force (unit tests)
plurality	most first places (ties split)	exact MILP	vs. brute force (unit tests)

Table 4: Spearman rank correlation of minimal L_1 across targets between rules, computed over targets feasible under both rules. Rules agree on which targets are cheap to push to first place; min is the outlier under gain10.

HELM MMLU, ceil cap

	mean	med.	min	Borda	Cond.
median	0.99				
min	0.98	0.94			
Borda	1.00	0.99	0.97		
Condorcet winner	0.99	1.00	0.95	0.99	
plurality	0.98	0.99	0.93	0.98	0.99

HELM MMLU, gain10 cap

	mean	med.	min	Borda	Cond.
median	0.92				
min	0.75	0.77			
Borda	0.99	0.96	0.77		
Condorcet winner	0.95	0.97	0.78	0.97	
plurality	0.90	0.94	0.67	0.92	0.98

OLLB, ceil cap (score rules only)

	mean	med.	min	Borda	Cond.
median	0.99				
min	0.96	0.95			
Borda	–	–	–	–	–
Condorcet winner	–	–	–	–	–
plurality	–	–	–	–	–

Use of LLMs. The method contains no learned or LLM component. The research itself (planning, coding, running the analysis, drawing the figures and writing this paper) was carried out by LLM-based agents working in separate roles on a human-specified objective, with independent reviewing roles; every number in the paper comes from the code in the delivery, and references were checked against their bibliographic records. Several related-work entries were read at the level of abstract or description only, which the text states where it matters.

Table 5: On the same models, the median and min rules cost 5–20 times less per task on 39 leaf tasks than on six benchmark aggregates, while the mean's cost per task falls only 1.3–1.6 times: n, not the number of models, makes the order-statistic rules cheap. Top: the 4506 (leaf tasks) and 4496 (aggregates) Open LLM Leaderboard models, raw accuracy, ceil cap; targets ranked 11–30 and 31–100 under each board's own mean rule (so the groups are not the same models on both boards), 20 and 70 sampled targets per board. L_1/n: median points per task [95% bootstrap CI over targets, 1000 resamples, seed 0]; K/n: median share of tasks raised; gain10 inf.: share of the group that cannot reach first place when each task may rise by at most 10 points. Bottom: random task subsets of the top-500 slice of the leaf board (60 draws; $n = 39$ is the full board, one subset), 5th-ranked target, ceil cap; median L_1/n [95% CI over draws] and the ratio of the rule's median to the mean's. Ordinal rules (Borda, Condorcet, plurality) were run only on the top-500 slice (Appendix Table 6).

		ranks 11–30 ($N = 20$)			ranks 31–100 ($N = 70$)		
rule	board	L_1/n	K/n	gain10 inf.	L_1/n	K/n	gain10 inf.
mean	6 benchmarks	4.9 [4.6, 5.4]	17%	0%	6.9 [6.7, 7.2]	17%	0%
	39 leaf tasks	3 [2.6, 3.6]	5%	0%	5.4 [5.3, 5.6]	8%	0%
geometric mean	6 benchmarks	3.6 [3.1, 4.1]	17%	0%	5.4 [5.2, 5.6]	17%	0%
	39 leaf tasks	1.7 [1.5, 1.9]	5%	0%	3.2 [3.1, 3.3]	8%	0%
median	6 benchmarks	4.4 [3.6, 4.7]	33%	80%	5.4 [5.2, 5.5]	33%	100%
	39 leaf tasks	0.22 [0.18, 0.26]	9%	0%	0.61 [0.57, 0.66]	13%	34%
min	6 benchmarks	2.4 [1.6, 2.7]	33%	45%	2.6 [2.4, 2.8]	33%	60%
	39 leaf tasks	0.27 [0.13, 0.51]	5%	30%	0.47 [0.32, 0.55]	8%	41%

L_1/n, 5th-ranked target	$n = 6$	$n = 24$	$n = 39$
mean	2.9 [2.2, 3.8]	1.8 [1.3, 2.7]	1.6
median	1.4 [0.96, 1.8]	0.31 [0.24, 0.53]	0.067
min	0.41 [0.31, 0.66]	0.12 [0.08, 0.15]	0.05
median / mean	0.48	0.18	0.04
min / mean	0.14	0.07	0.03

Table 6: On the top-500 slice of the 39-leaf-task board the Borda and Condorcet requirements are among the cheapest rules per task, and plurality is infeasible for most targets under gain10. Same conventions as Table 5: raw accuracy, ceil cap, targets ranked 11–30 and 31–100 under the slice's own mean rule (20 and 43 targets), L_1/n as median [95% bootstrap CI over targets], K/n the median share of tasks raised, gain10 inf. the share of targets that cannot reach first place with at most 10 points per task. The ordinal rules were run only on this slice. The Condorcet winner is an upper bound for Copeland and Schulze.

	ranks 11–30 ($N = 20$)			ranks 31–100 ($N = 43$)		
rule	L_1/n	K/n	gain10 inf.	L_1/n	K/n	gain10 inf.
mean	3 [2.6, 3.6]	5%	0%	5.1 [4.9, 5.4]	8%	0%
geometric mean	1.7 [1.4, 1.9]	5%	0%	3 [2.9, 3.3]	8%	0%
harmonic mean	1.1 [1, 1.3]	5%	0%	2 [1.9, 2.1]	8%	0%
median	0.22 [0.19, 0.27]	9%	0%	0.61 [0.54, 0.66]	13%	35%
min	0.27 [0.13, 0.5]	5%	30%	0.48 [0.35, 0.57]	8%	42%
Borda	0.86 [0.75, 1.2]	13%	0%	1.8 [1.7, 1.9]	28%	0%
Condorcet winner	0.75 [0.55, 0.81]	21%	5%	1.5 [1.5, 1.7]	33%	37%
plurality	2 [1.7, 2.2]	21%	80%	2.8 [2.6, 3]	21%	100%

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Table 7: The rules that are cheapest with optimal targeting are the ones that gain most from it: a random split of the budget needs about 8–40 times the optimal budget under the median, min and plurality rules, and nothing extra under the mean. HELM, targets ranked 2–20 with optimal $L_1 \geq 0.05$ points, 100 random allocations per target. Spread factor: smallest multiple of the optimal L_1 at which a random split of the budget over all 57 tasks makes the target first, median over the N targets of the per-target median factor, with the 25%–75% range over targets. K^* random: share of targets for which choosing at random only as many tasks as the optimum raises, even with every chosen task raised to the cap, cannot reach first place in the median draw. Condorcet winner (a sufficient but not necessary condition for first place under Copeland and Schulze) is an upper bound for those rules.

	ceil			gain10		
rule	spread factor	N	K^* random: inf.	spread factor	N	K^* random: inf.
mean	1.00 [1.00, 1.00]	20	95%	1.00 [1.00, 1.00]	20	90%
geometric mean	1.16 [1.15, 1.24]	19	100%	1.09 [1.05, 1.17]	18	100%
harmonic mean	1.35 [1.33, 1.5]	19	100%	1.19 [1.10, 1.35]	18	100%
Borda	2.9 [2.7, 3.4]	20	100%	2.4 [2.0, 3.5]	18	100%
Condorcet winner	4.0 [3.5, 5.3]	19	100%	3.7 [3.2, 5.5]	19	100%
plurality	7.7 [7.4, 10.6]	18	100%	7.7 [7.1, 11.3]	18	100%
median	12.2 [8.9, 17.0]	19	95%	12.2 [8.4, 17.7]	19	100%
min	38.4 [26.4, 68.4]	19	100%	43.4 [26.6, 70.7]	11	100%

Table 8: On HELM the Condorcet-winner requirement costs the same as exact Copeland, on the six-task OLLB slice it overstates Copeland's cost 1.4–8 times. Exact Copeland minimum investment (MILP) against the Condorcet-winner (CW) requirement, 34 target×cap cases with ranks as shown; ratio of the CW to the Copeland L_1 over the cases feasible for CW; the two HELM gain10 cases infeasible for CW (ranks 50, 70) are also infeasible for Copeland. Copeland's cost is below CW's in every case, as it must be.

dataset	cap	cases	ranks	CW/Copeland L_1	CW infeas.	Copeland $\leq$ CW
OLLB top-500	ceil	5	2–11	1.40–8.06	0	yes
OLLB top-500	gain10	5	2–11	1.40–8.06	0	yes
HELM MMLU	ceil	12	2–70	1.000–1.001	0	yes
HELM MMLU	gain10	12	2–70	1.000–1.001	2	yes

Justification: The abstract and Section 1 state the question, the model (cap-limited minimum investment), the closed forms (Section 4) and the empirical findings (Section 5), including that results are conditional on the headroom cap and on two leaderboards, and that infeasibility is a result.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
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- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 6 lists modelling assumptions (reachability, static rivals, one target, cost in points not compute) and limits of the evidence (small HELM rank groups, sampled ordinal runs, one snapshot, observational empirical caps, unrun rules).

Guidelines:

Table 9: HELM's rank-2 and rank-3 targets are within evaluation noise of the leader, so their investment costs are not a statement about rules. Gap in the 57-subject mean score between the leader (Claude 3.5 Sonnet 20241022) and each of the next ten models, in points and in binomial standard errors of the gap (independent cell errors; models share items, so the paired SE is smaller and this ratio is a lower bound on the gap in paired SE units); last column: minimal L_1 of the target under the mean rule (ceil).

rank	target	gap to leader (points)	gap / SE	mean-rule L_1 (points)
2	DeepSeek v3	0.07	0.1	4.0
3	Gemini 1.5 Pro (002)	0.39	0.8	22.5
4	Claude 3.5 Sonnet (20240620)	0.73	1.5	41.8
5	Claude 3 Opus (20240229)	2.70	5.4	154.0
6	Llama 3.1 Instruct Turbo (405B)	2.74	5.5	156.3
7	GPT-4o (2024-08-06)	2.94	5.9	167.6
8	GPT-4o (2024-05-13)	3.10	6.2	176.8
9	Qwen2.5 Instruct Turbo (72B)	3.90	7.7	222.4
10	Gemini 1.5 Pro (001)	4.56	9.0	260.2
11	GPT-4 (0613)	4.84	9.4	275.7

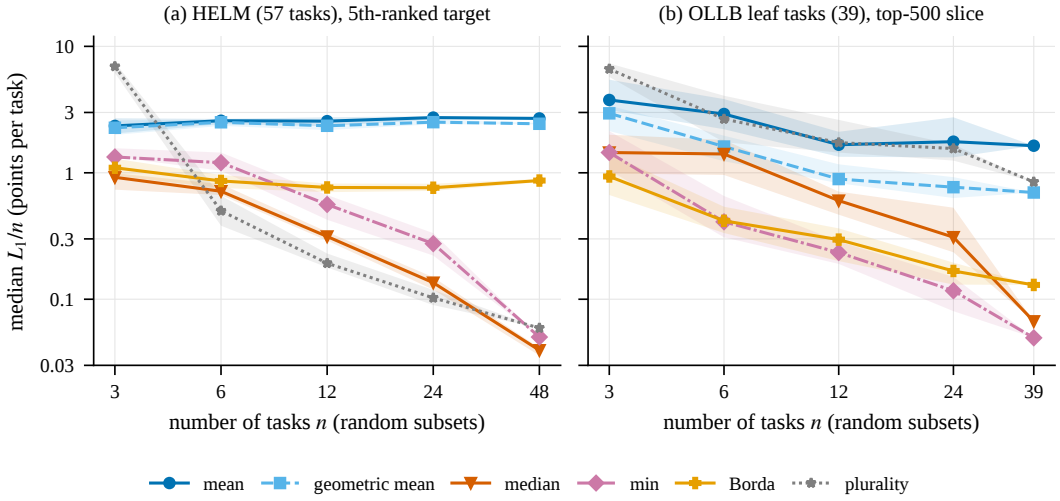


Figure 4: Per-task cost of the mean rule is flat in n on HELM and falls slowly on the leaf board, while the median, min and plurality rules' costs fall steeply: absolute levels behind the ratios of Figure 3c, d. Median L_1/n (points per task, ceil cap, log axis) over random subsets of n tasks for the target ranked 5th under each rule's own ranking; (a) HELM, $n \leq 48$ of 57 tasks, 200 draws; (b) top-500 slice of the 39-task leaf board (raw accuracy), 60 draws, $n = 39$ one subset. Bands are 95% CIs over draws.

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
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- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
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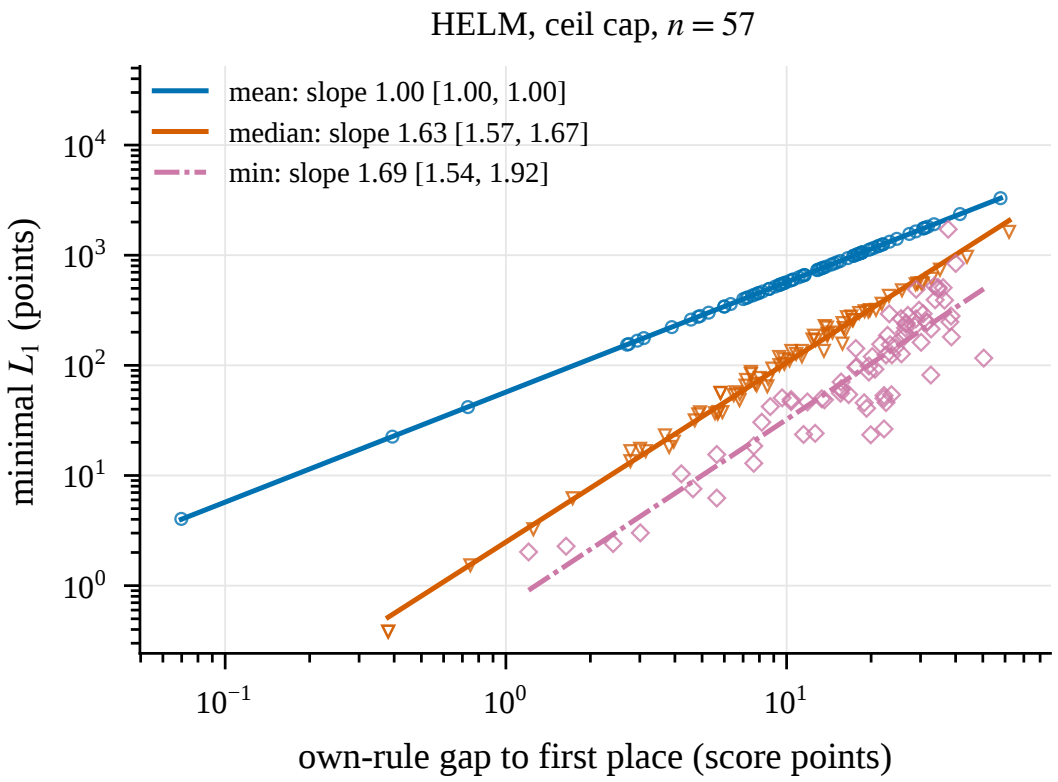


Figure 5: Under the mean rule the minimal investment is proportional to the gap to first place (slope 1), while the median and min costs grow faster than the gap. HELM, ceil cap, 78 targets: minimal L_1 against the target's gap to first place under its own rule, log–log; markers are targets, lines are least-squares fits of $\log L_1$ on $\log$ gap with the slope and its 95% interval from `code/results/tables/elasticity_helm.csv` (targets with zero cost or zero gap excluded).

used reliably to provide closed captions for online lectures because it fails to handle technical jargon.

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3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

   Justification: Propositions 1–4 (Section 4) state their assumptions; proofs are in Section 4 or Appendix A, and the closed forms are also checked numerically against brute force and the MILP solvers (Section 5.2).

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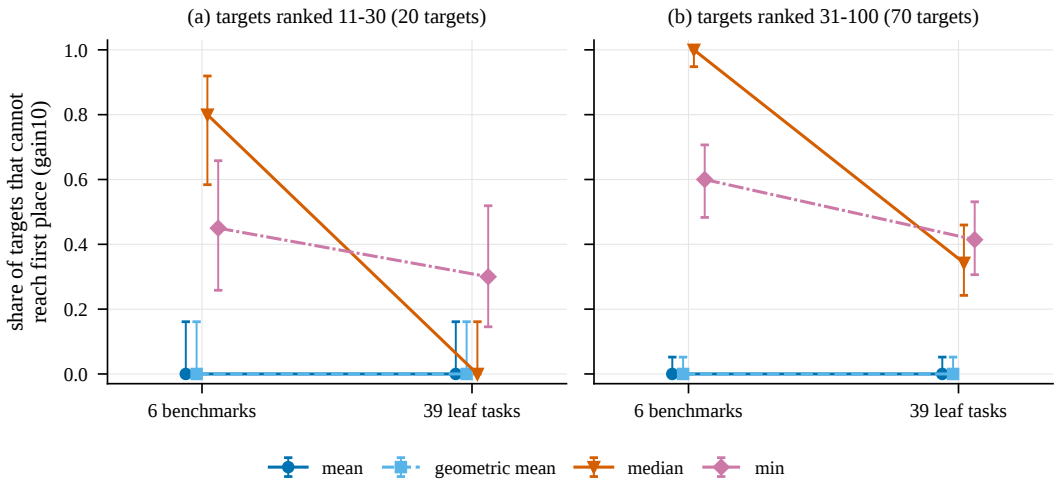


Figure 6: On the finer board the median and min rules become feasible for more targets under the gain10 cap. Share of targets that cannot reach first place with each task's gain capped at 10 points, for the same models scored on six aggregates and on 39 leaf tasks (targets ranked 11–30, 20 targets, and 31–100, 70 targets, under each board's own mean rule); bars are 95% Wilson intervals. Score rules only: the ordinal rules were not run on the full leaf board.

- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
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   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Section 3 defines the problem, caps, rules and solvers; Section 5.1 gives data versions, filtering, grouping and bootstrap settings; Appendix B and the delivered `REPRODUCE.md` give versions, checksums, seeds and commands.

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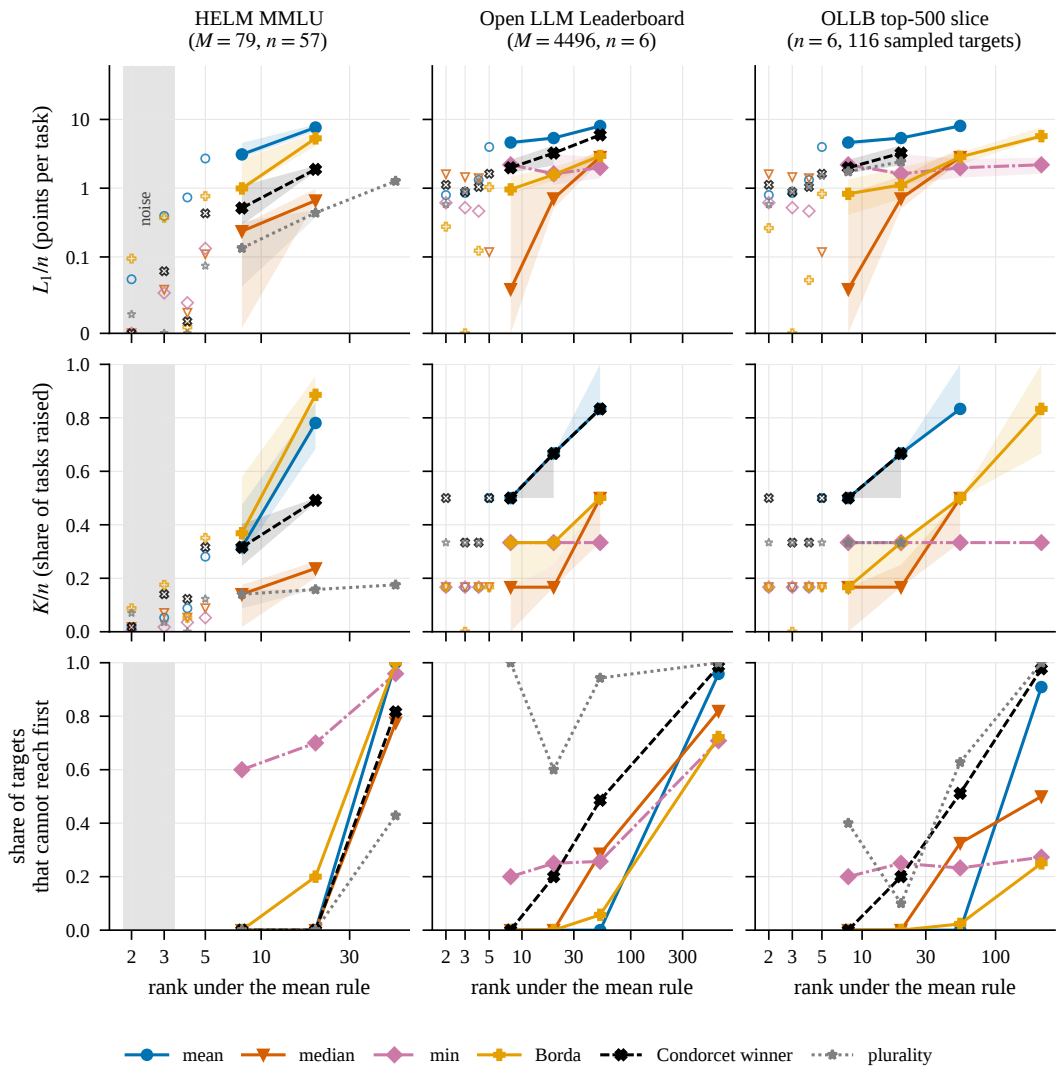


Figure 7: With a bound of 10 points per task, first place becomes unreachable for most low-ranked targets, and the cost curves stop where more than half of a group is infeasible. Same layout as Figure 1 for the gain10 cap (scores may rise by at most 10 points); the bottom row shows the share of the targets in each rank group that cannot reach first place. OLLB (middle) compares all rules on the targets common to all rules run under this cap (rank-stratified sample); the right panel is the top-500 slice. Medians, bands and hollow markers as in Figure 1. Condorcet winner (a sufficient but not necessary condition for first place under Copeland and Schulze) is an upper bound for those rules.

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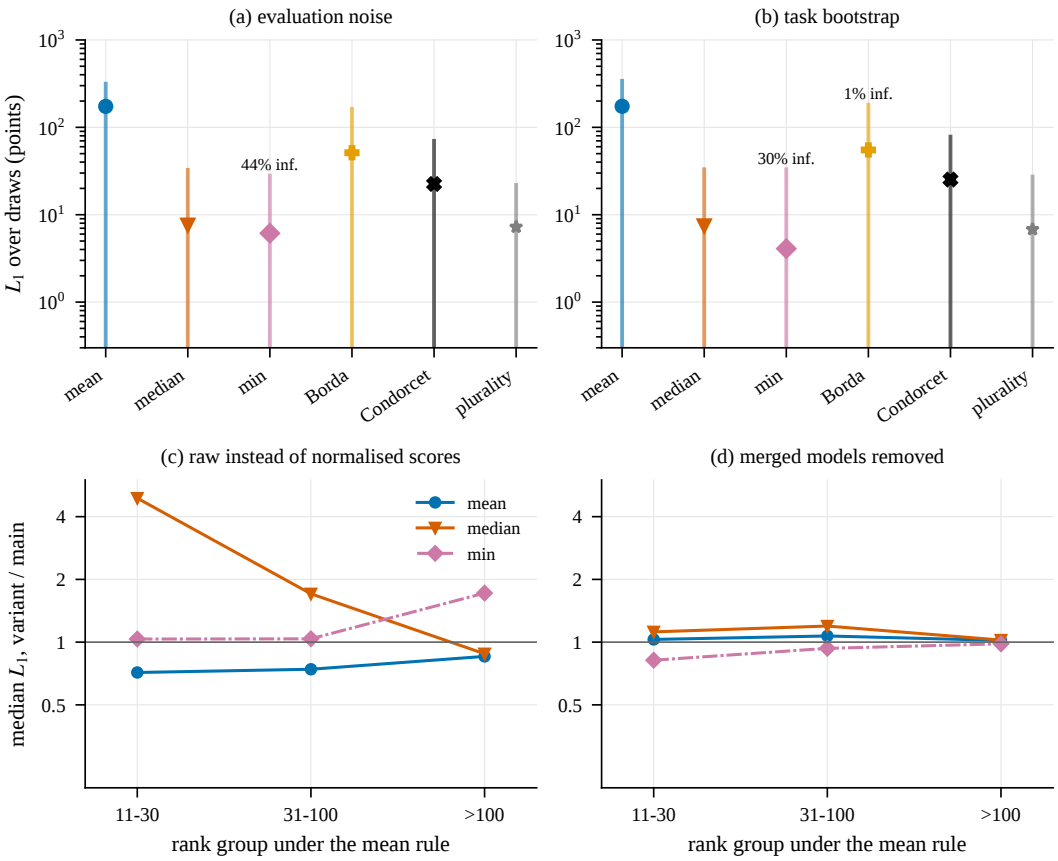


Figure 8: Rules that are cheap on the observed scores mostly stay cheap under evaluation noise and task resampling, except min, whose cost is unstable. (a, b) HELM, gain10 cap, targets ranked 2–11, 100 Monte-Carlo draws: (a) evaluation noise added to every score (binomial standard error), (b) bootstrap over tasks. Dot = median L_1 over draws and targets, bar = 2.5–97.5% range of the draws (lower end truncated at 0.3 points; it reaches 0 where the target is first in some draws); labels give the share of infeasible draws. The min rule's cost is the least stable. (c, d) OLLB, ceil cap, all targets: ratio of the median L_1 of a variant to that of the main run, per rank group of the target under the mean rule in that variant; (c) raw instead of normalised scores, (d) merged models removed (3789 of 4496 targets remain).

(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Both data sets are public (versions and SHA-256 in `REPRODUCE.md`); code, a Makefile and the script-to-table map are in the anonymous delivery (`code/`, `scripts/`, `REPRODUCE.md`).

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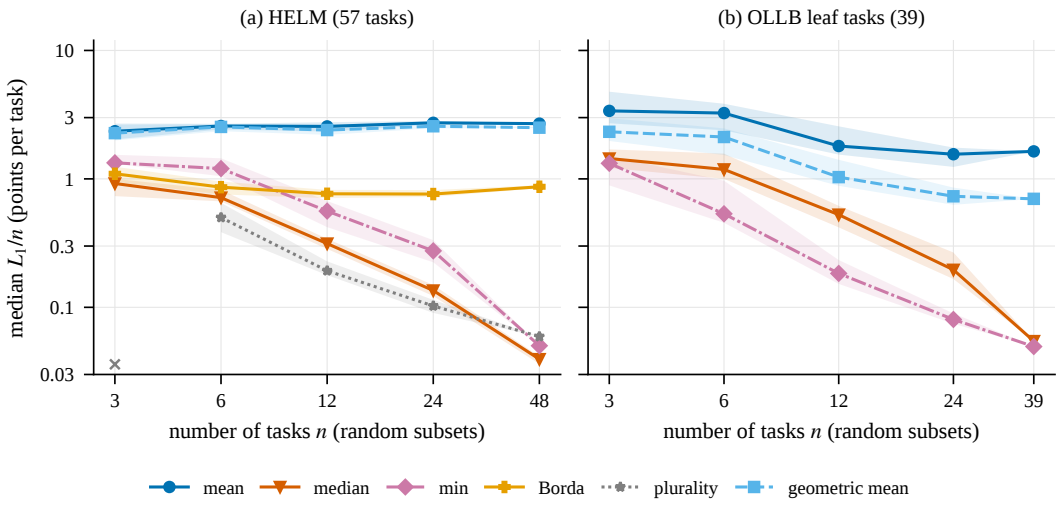


Figure 9: Under the gain10 cap the cost per task also falls with the number of tasks for order-statistic rules, with some subsets infeasible. Median L_1/n of the target ranked 5th under each rule's own ranking against n, for random task subsets of HELM (a, 200 draws) and of the 39-task OLLB leaf board (b, 100 draws; $n = 39$ is the full board); bands are 95% CIs over draws. A cross on the lower axis marks an n at which more than half of the draws are infeasible (plurality, HELM, $n = 3$); points with some infeasible draws are medians over all draws, with infeasible draws counted as infinite cost. Borda and plurality were not run on the leaf board under gain10.

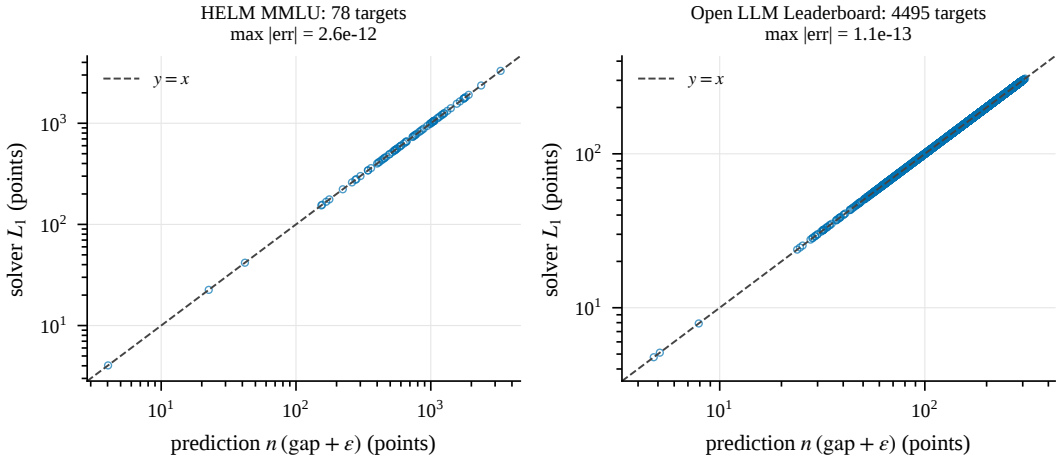


Figure 10: The closed form for the mean rule matches the exact solver on every target: minimal L_1 equals $n(\text{gap} + \varepsilon)$ with $\varepsilon = 10^{-3}$ at the ceil cap. Each marker is one target (HELM, $n = 57$: 78 targets; OLLB, $n = 6$: 4495 targets); dashed line $y = x$; the largest absolute deviation is given above each panel.

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6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: There is no training: the setting is the data filtering, caps, margin $\varepsilon = 10^{-3}$, rank groups, stratified samples, solver tolerances and seeds, in Sections 3 and 5.1 and Appendix B.

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7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: Group medians carry 1 000-resample bootstrap percentile intervals over targets (seed 0); Monte-Carlo evaluation-noise and task-bootstrap runs are in Section 5.5. Rank groups with few targets (HELM) have wide or degenerate intervals, which is stated.

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8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: CPU only (2 CPUs, 4 GB); the full pipeline takes 3.75 h wall back to back (about 3–4 CPU-hours), with per-stage times in Appendix B and `REPRODUCE.md`; development runs are not included in that figure.

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Justification: Section 6 discusses that the analysis could in principle inform targeted training for leaderboard gaming and that its main use should be leaderboard design; the model is coarse and uses no model identity.

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